

Purdue University

# Buck Converter and Boost Converter Design

ECE 61016: Power Electronic Converters and Systems

**Majharul Hoque**

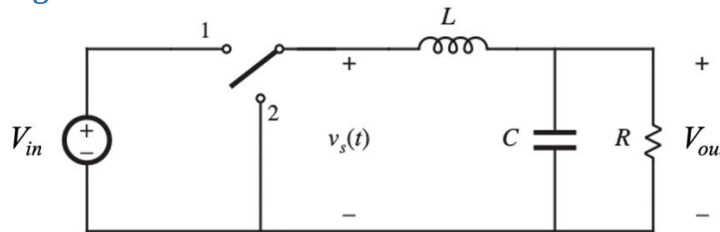
# ECE 61016: Power Electronic Converters and Systems

Purdue University

Project #1

Name: Majharul Hoque

## 1. A buck converter is given below:



(a) The buck converter above has an input voltage of 12 V and the required output voltage is  $V_{out} = 1$  V for a GPU at nominal output current of 10 A. The switching frequency is at 300 kHz. Design LC filter that meets:

- Inductor ripple current < 30% of the output current
- Capacitor voltage ripple < 5% of the output voltage
- Cutoff frequency < 300 kHz

The switching period,  $T = \frac{1}{F_s} = \frac{1}{300\text{kHz}} = 3.33 \mu\text{s}$

Duty cycle for a buck converter,

At  $V_{in} = 10\text{V}$

$$D_{10} = \frac{\langle V_0 \rangle}{V_{in}} = \frac{1}{10} = 0.1$$

At  $V_{in} = 12\text{V}$

$$D_{10} = \frac{\langle V_0 \rangle}{V_{in}} = \frac{1}{12} = 0.0833$$

The reqd. inductance for 30% inductor current ripple :

Typically, peak-to-peak inductor current ripple is given as a percentage of the output current.

So,  $\Delta i_L = 30\% \times i_0 = 3\text{A}$

Then, inductance required,  $L = \frac{V_{in} - \langle v_0 \rangle}{\Delta i_L} \times DT = \frac{12 - 1}{3} \times (0.0833 \times 3.33) \geq 1.01 \mu\text{H}$

The reqd. capacitance for capacitor voltage ripple < 5% of the output voltage:

Then, capacitance required,  $C = \frac{D \times i_0}{F_s \Delta V_0} = \frac{10 \times 0.0833}{300\text{kHz} \times 0.05} = 55 \mu\text{F}$

See the schematic below for an LTSPICE implementation of the buck converter, that matches the requirements put- forward:

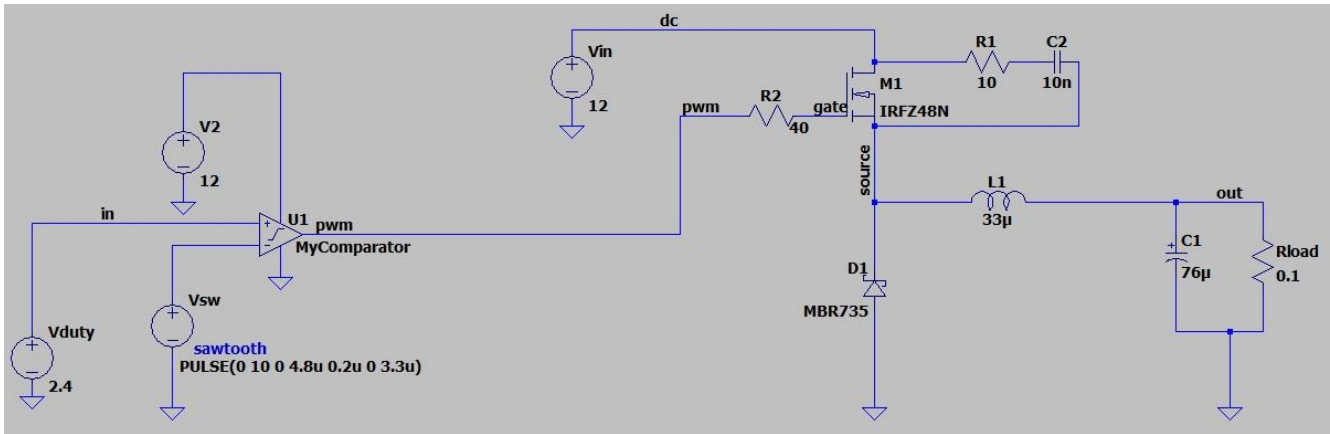


Figure 1: An LTSPICE implementation of Buck Converter. An inhouse comparator was used to generate a Pulse signal for the NMOS MOSFET. The Vduty voltage source is corresponding to the duty cycle. The inductor and capacitance values are what was practical for the requirement. The load resistance for buck converter is usually low.

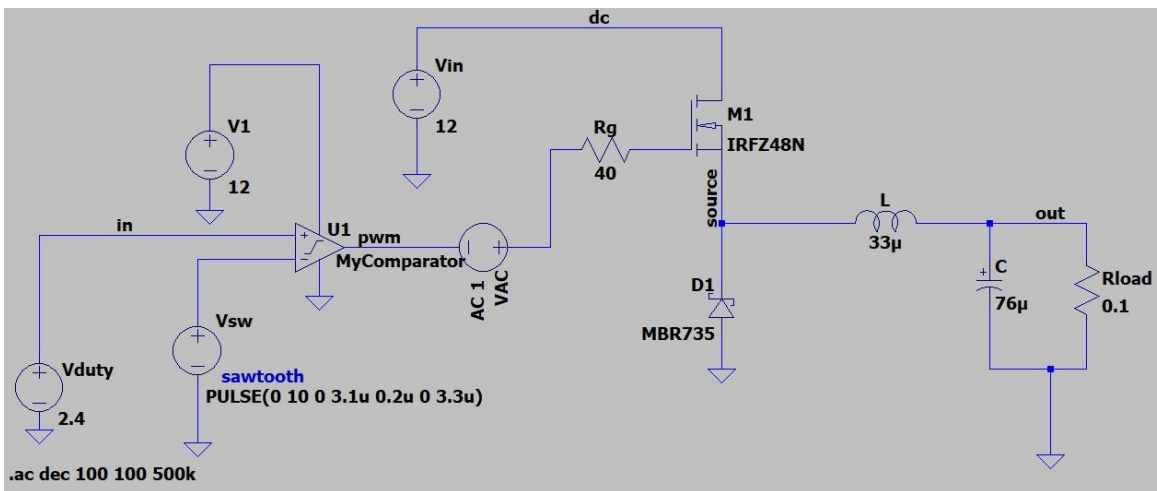


Figure 2: Openloop buck converter circuit for Control transfer function.



Figure 3: Openloop buck converter magnitude-phase for Control transfer function.

- (b) Pick appropriate Si MOSFET and/or diode (use of two MOSFETs is recommended) from the list given in LTspice device list. Plot the input, output, voltage, current, and input gate voltage of the upper switch. In this simulation, use ideal inductor and capacitor without any resistance. Estimate the total converter efficiency.

Below schematic is generated using LTSPICE for Open loop buck converter simulation.

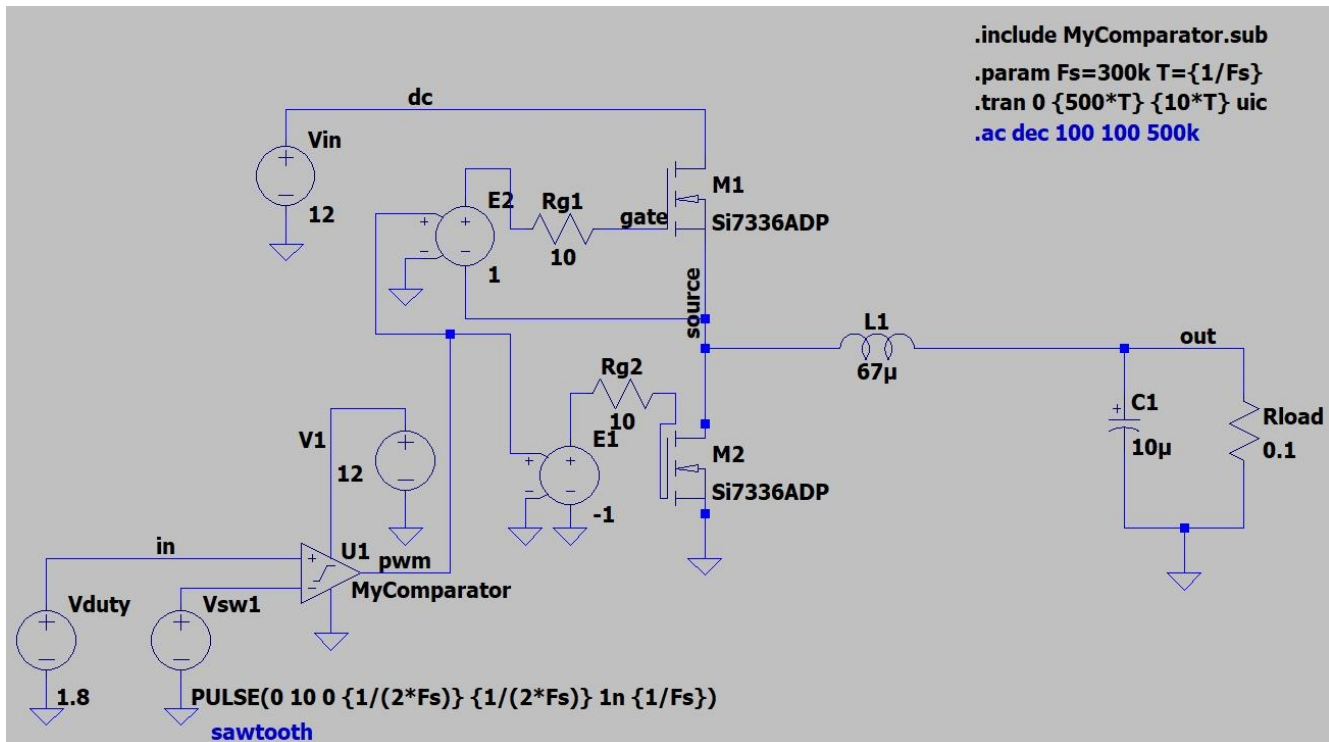


Figure 4: Implementation of the Open Loop schematic of the Buck converter using two MOSFETs. The PULSE wave is generated using the SAWTOTH waveform and Vduty voltage source. Vduty voltage source is fine tuned to generate the output voltage and current required.

The Si MOSFETs used are Si7336ADP from Vishay Siliconix . Some of the important datasheet parameters have been given below with plots from the datasheet:

| MOSFET Specification | Part No.   | Manufacturer     | $V_{ds}$ | $R_{ds}$                                                | $Q_g$                                                        |
|----------------------|------------|------------------|----------|---------------------------------------------------------|--------------------------------------------------------------|
| Values               | Si7336ADP* | Vishay Siliconix | 30V      | 0.0040 $\Omega$ (max at $V_{GS} = 4.5$ V, $I_D = 19$ A) | 50nC (max at $V_{DS} = 15$ , $V_{GS} = 4.5$ V, $I_D = 20$ A) |

|  | Input voltage (RMS) | Output voltage (RMS) | Input current (RMS) | output current (RMS) | Duty ratio [%] | Efficiency [%] |
|--|---------------------|----------------------|---------------------|----------------------|----------------|----------------|
|  |                     |                      |                     |                      |                |                |

|        |     |         |         |         |               |     |
|--------|-----|---------|---------|---------|---------------|-----|
| Values | 12V | 1.0448V | 4.5553A | 10.429A | 0.0833 to 0.1 | 39% |
|--------|-----|---------|---------|---------|---------------|-----|

\* See Reference for Datasheet

Plots:

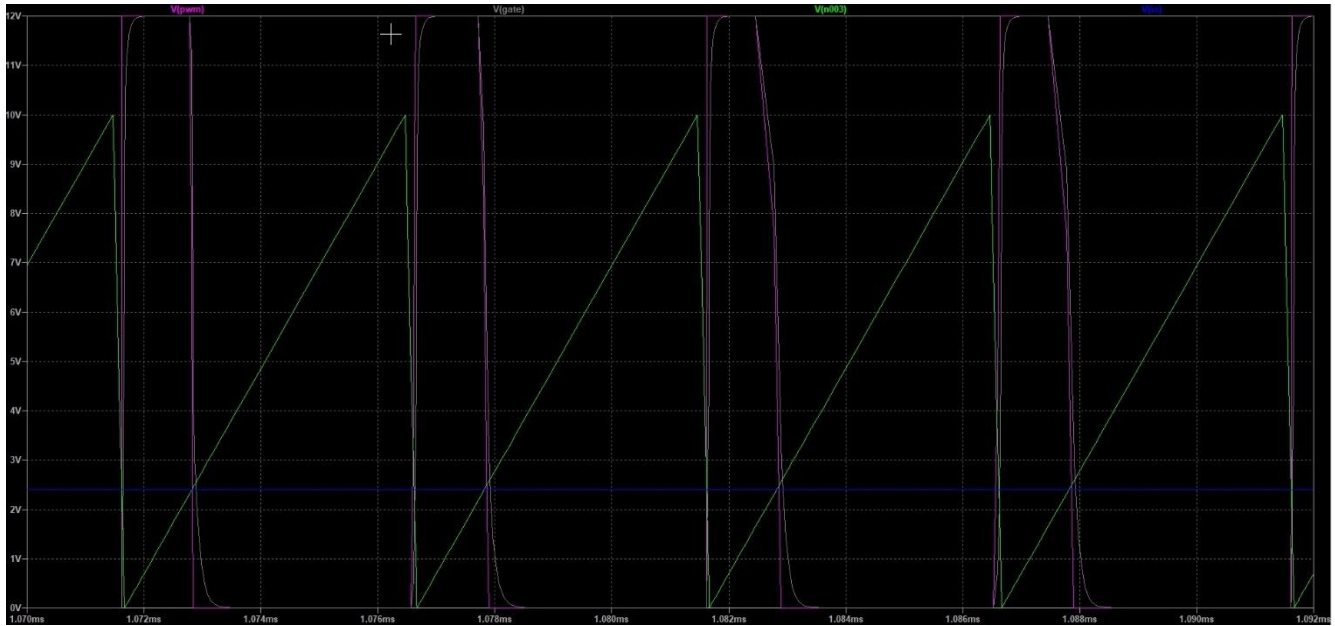


Figure 5: The input output plots. PWM signal in pink, Gate voltage in grey, Sawtooth waveform in green and input voltage in blue.

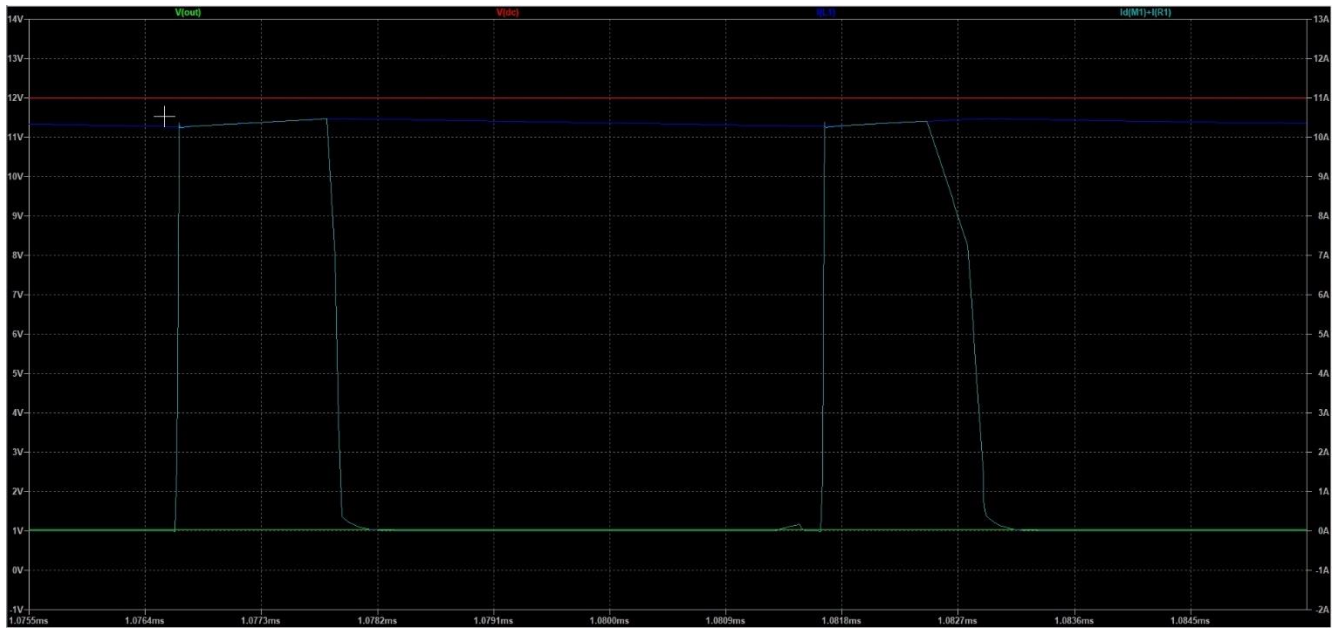


Figure 6: Vdc is the input DC voltage(red),input current through the upper switch MOSFET drain, current through the inductor, and output voltage all values are in steady state.

(c) Add 0.1 ohm of the equivalent series resistance (ESR) to the inductor. Plot the input, output, voltage, current, and input gate voltage of the upper switch. In this simulation, use ideal capacitor without any resistance. Estimate the total converter efficiency.

|        | Input voltage (RMS) | Output voltage (RMS) | Input current (RMS) | output current (RMS) | Duty ratio [%] | Efficiency [%] |
|--------|---------------------|----------------------|---------------------|----------------------|----------------|----------------|
| Values | 12                  | 1.0225V              | 1.7493A             | 3.4082A              | 0.0833 to 0.1  | 16.6%          |

Adding inductor internal resistance  $ESR = 0.1\Omega$  in series will significantly reduce the output current.

Plots:

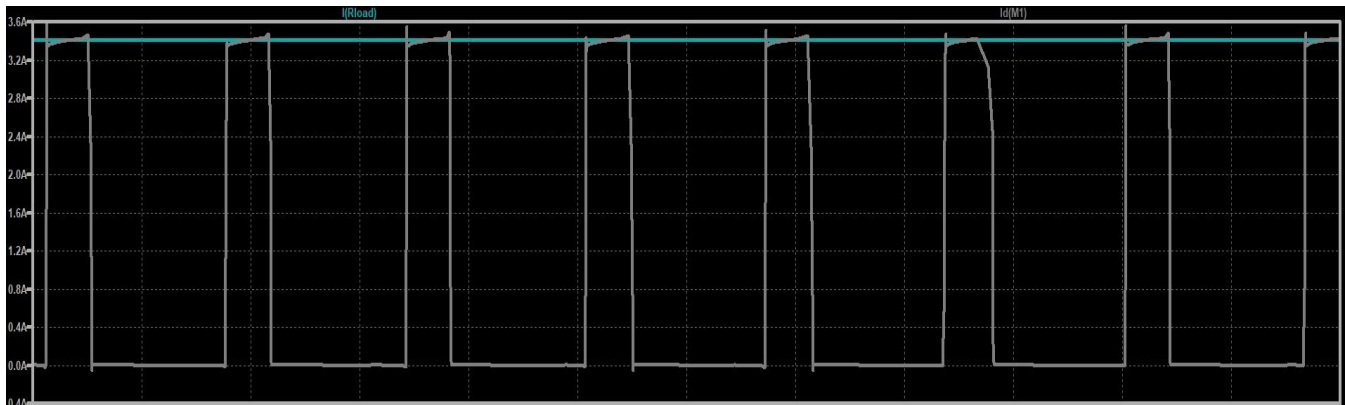


Figure 7: input output current plots; output current in electric blue, input current in grey.

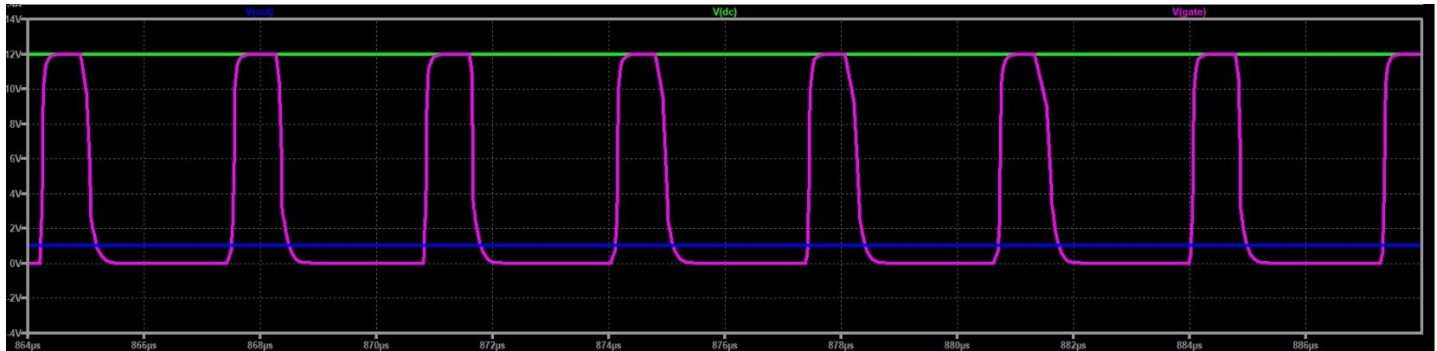


Figure 8: input output voltage plots; Input voltage(green), output voltage(blue), gate voltage of the upper MOSFET switch(magenta).

(d) Calculate switching losses (turn on and off) and compare them with simulation results.

See the APPENDIX-A for detailed formula for the calculation. Here I have summarized the calculated values:

| Switching times        | Calculated values | Datasheet values <sup>^</sup> |
|------------------------|-------------------|-------------------------------|
| $T_{d(ON)}$            | 0.893 nS          | 12-24nS                       |
| rise time, $t_R = t_3$ | 1.68nS            | 10-20nS                       |

|                                 |                               |         |
|---------------------------------|-------------------------------|---------|
| $T_{d(OFF)}$                    | 4.84nS                        | 30-60nS |
| Fall time, $t_f = t_{VR} = t_3$ | 1.83nS (for $R_G = 1\Omega$ ) | 10-20nS |

^see reference section for datasheet(Si7336ADP)

| Losses        | Calculated value | Simulation result |
|---------------|------------------|-------------------|
| Turn on loss  | 0.04266 Watts    | 1.1297 $\mu$ J/S  |
| Turn off loss | 0.033 Watts      | 5.6276 $\mu$ J/S  |

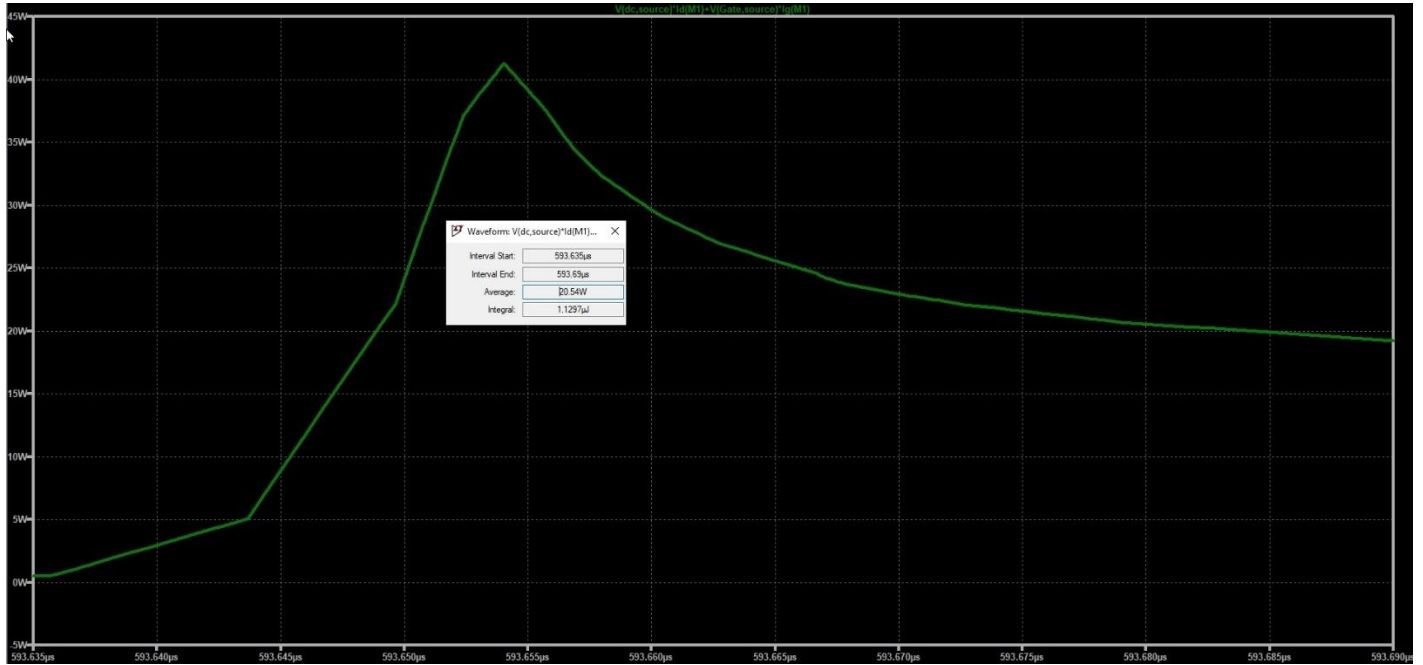


Figure 9: MOSFET turn-ON loss plot.

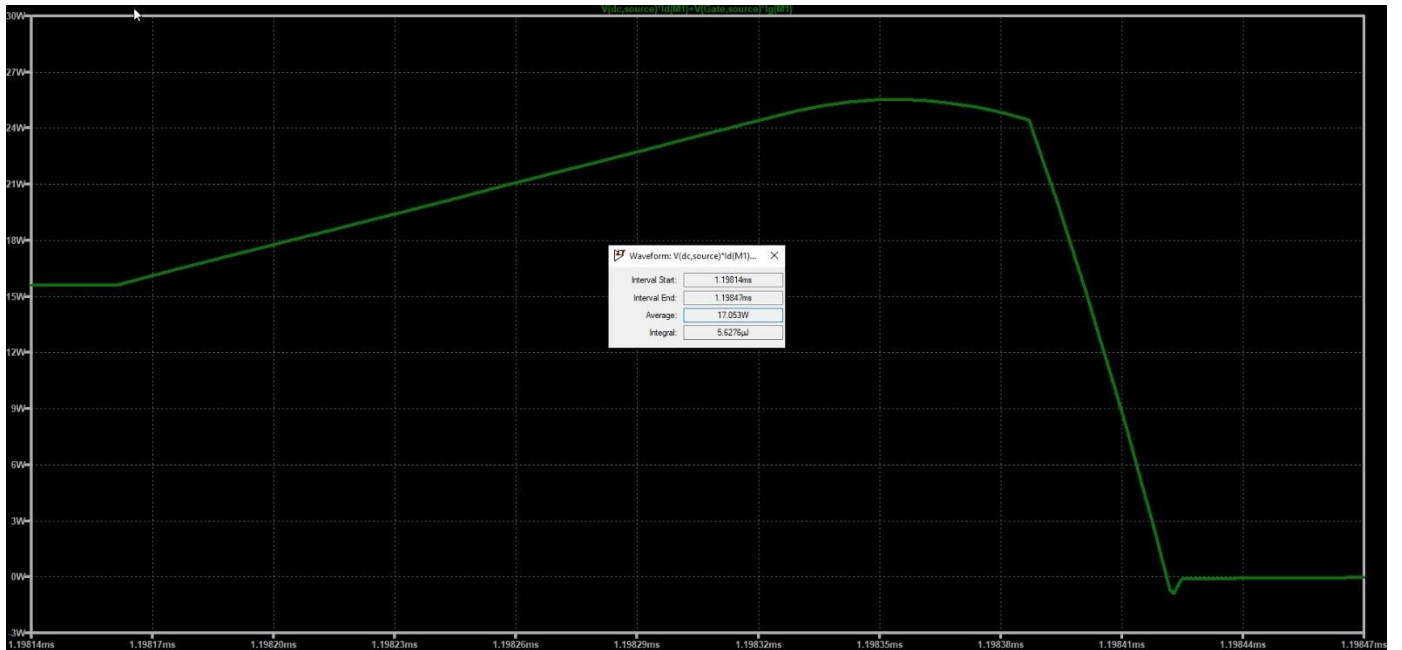


Figure 10: MOSFET turn-OFF loss plot.



(e) Implement analog closed loop controller (e.g., PI regulator) with measured voltage feedback.

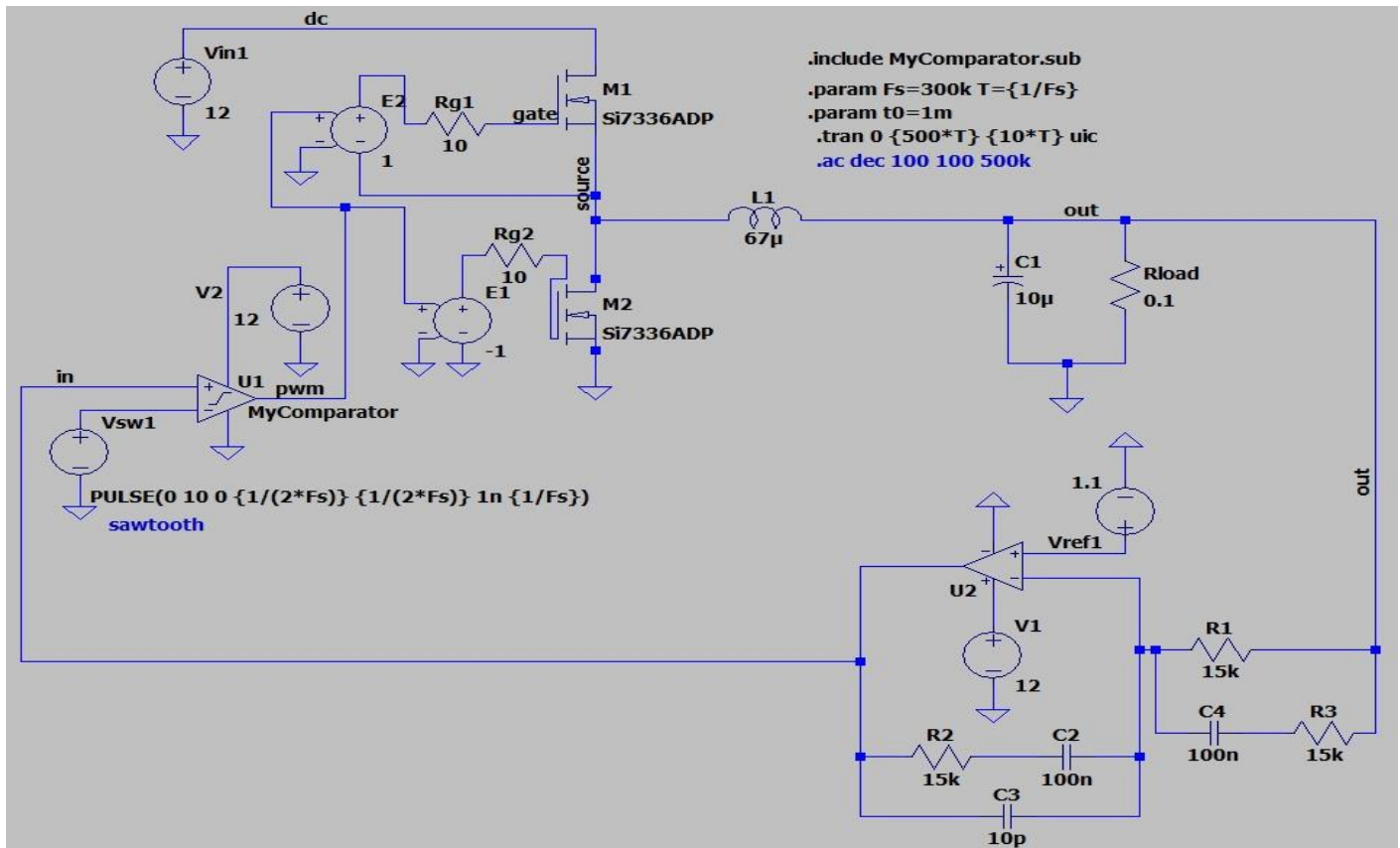


Figure 11: Closed loop implementation of Buck Converter with PI regulator and feedback configuration

(f) Compare closed loop system loop gain magnitude and phase with analytically calculated small signal model loop gain magnitude and phase.

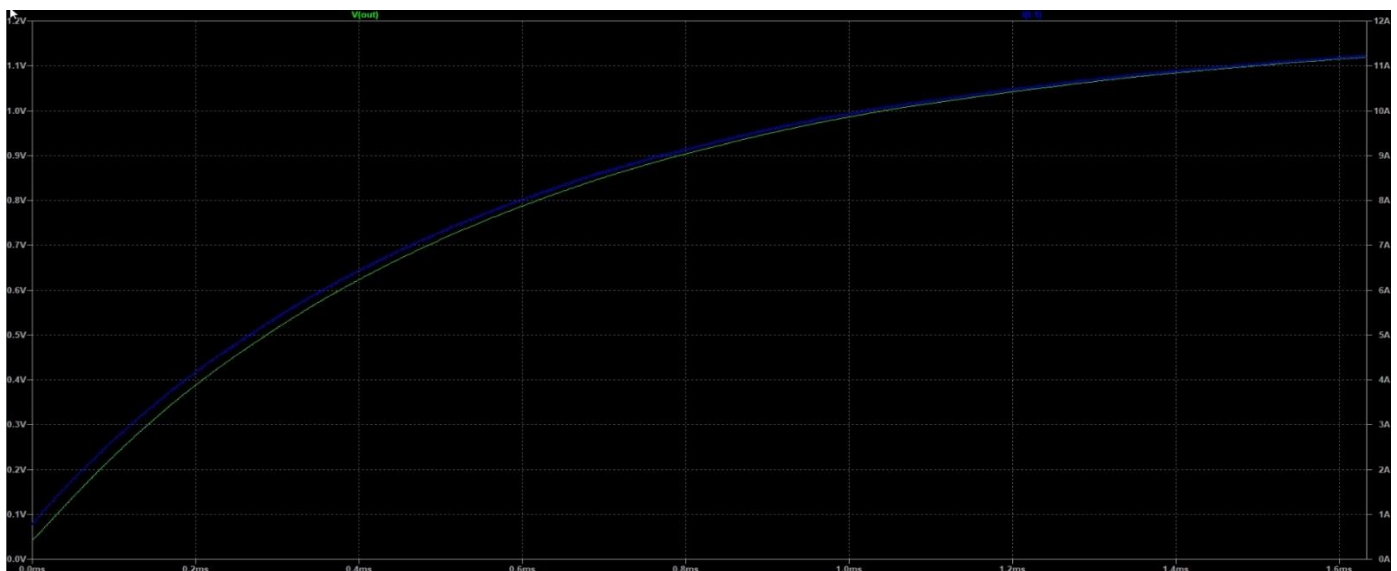
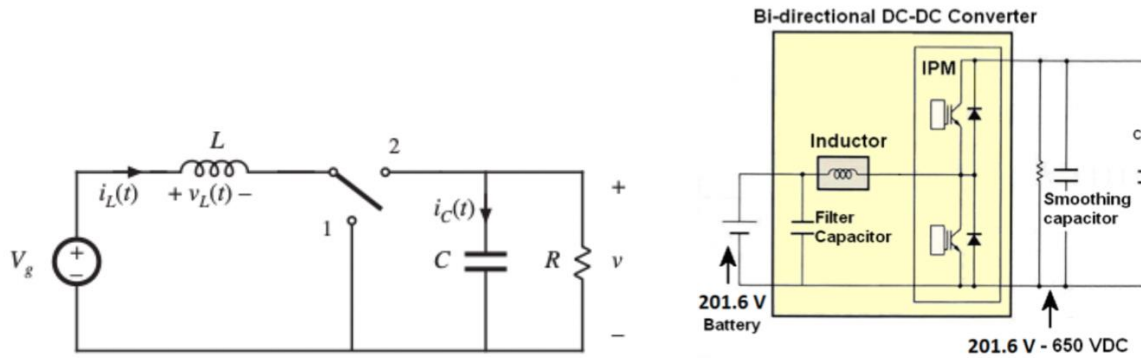


Figure 12: Closed Loop output voltage current plot



2. A boost converter is used in hybrid electric vehicle for boosting the input battery voltage from 150 V to 600 V. The converter switching frequency is at 5 kHz.



- (a) The boost converter above has an input voltage of 150 V and the required output voltage is  $V_{out} = 600$  V for a laptop USB charger at nominal output current of 15 A. The switching frequency is at 5 kHz. Design LC filter that meets:

- Inductor ripple current < 40% of the output current
- Capacitor voltage ripple < 5% of the output voltage
- Cutoff frequency < 5 kHz

The switching period ,  $T = \frac{1}{F_s} = \frac{1}{5\text{kHz}} = 200\mu\text{s}$

Duty cycle for a boost converter,

At  $V_{in} = 150\text{V}$

$$D_{10} = 1 - \frac{\langle V_{in} \rangle}{V_o} = 0.75$$

At  $V_{in} = 600\text{V}$

The reqd. inductance for 30% inductor current ripple :

Typically, peak-to-peak inductor current ripple is given as a percentage of the output current.

$$\text{So, } \Delta i_L = 40\% \langle i_o \rangle = 6\text{A}$$

Then, inductance required ,  $L = \frac{V_{in} D}{\Delta i_L F_s} \geq 3.75\text{mH}$

The reqd. capacitance for capacitor voltage ripple < 5% of the output voltage:

$$\text{Then, capacitance required, } C = \frac{D \langle i_o \rangle}{F_s \Delta V_o} = \frac{10 \times 0.0833}{300\text{kHz} \times 0.05} = 75\mu\text{F}$$

See the schematic below for an LTSPICE implementation of the boost converter, that matches the requirements put- forward:

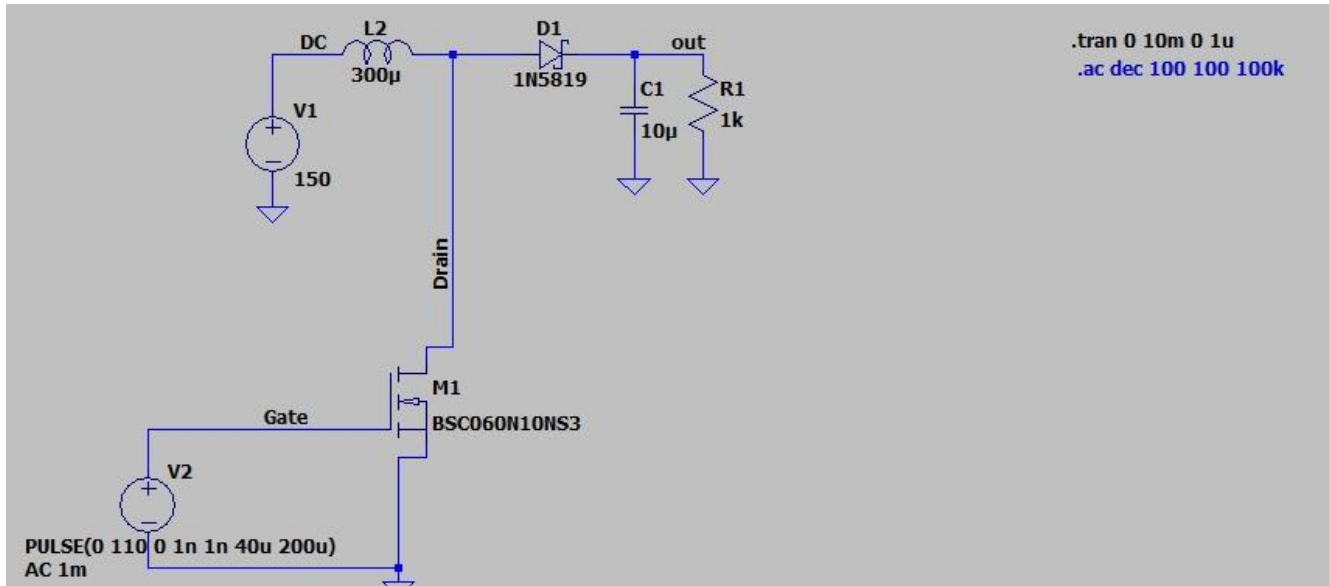


Figure 13: Boost converter design with PWM pulse signal.

**(b) Implement IGBTs given in D2L (AUIRGDC0250). Plot the input, output, voltage, current, and input gate voltage of the upper switch. In this simulation, use ideal inductor and capacitor without any resistance. Estimate the total converter efficiency.**

I couldn't use IGBT but here is a replacement PMOS MOSFET instead.

| MOSFET Specification | Part No.     | Manufacturer | $V_{ds}$ | $R_{ds}$                                                   | $Q_g$ |
|----------------------|--------------|--------------|----------|------------------------------------------------------------|-------|
| Values               | BSC060N10NS3 | infineon     | 100V     | 11.5 m $\Omega$<br>(max at $V_{GS} = 4.5$ V, $I_D = 19$ A) | 68nC  |

|        | Input voltage (RMS) | Output voltage (RMS) | Input current (RMS) | output current (RMS) | Duty ratio [%] | Efficiency [%] |
|--------|---------------------|----------------------|---------------------|----------------------|----------------|----------------|
| Values | 150V                | 525.35V              | 6.3388A             | 15A                  | 0.75           | 25%            |

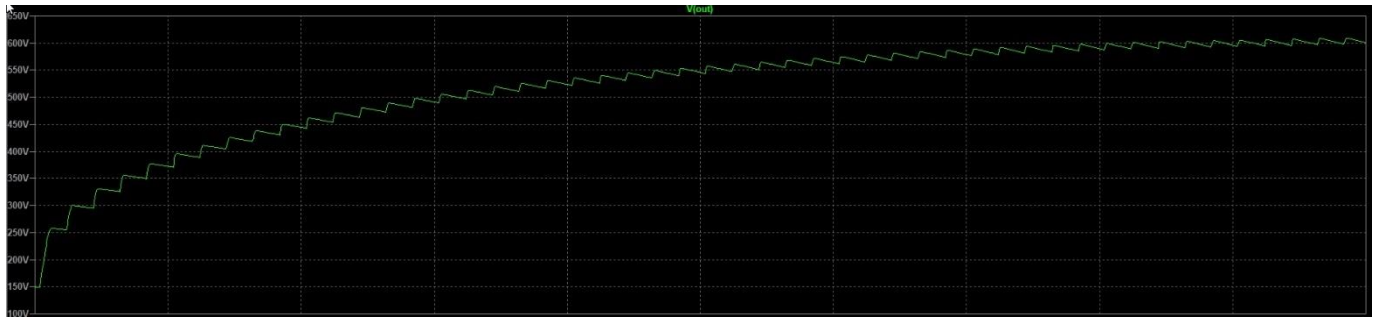


Figure 15: Boost converter output Voltage goes up to 600V.

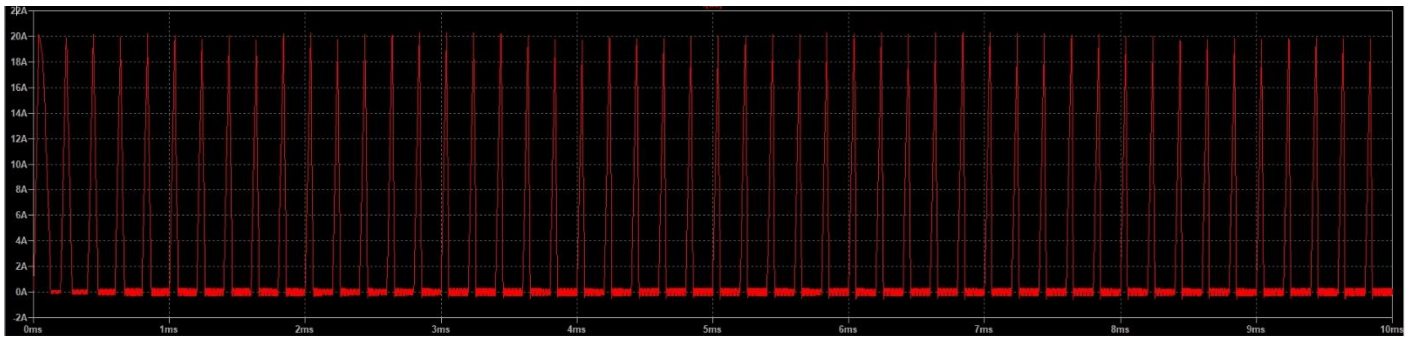


Figure 14: Boost converter Current through inductor with 5A ripple current

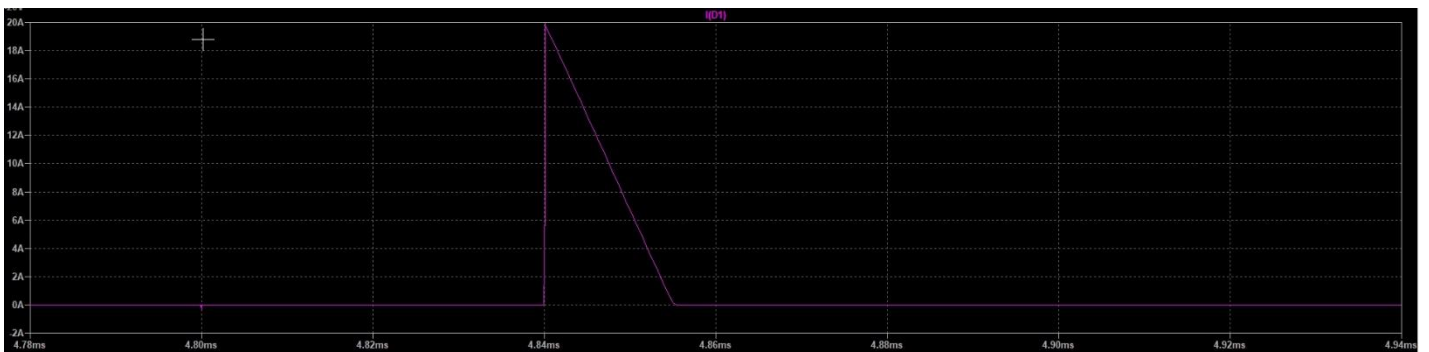


Figure 16: current through the inductor.

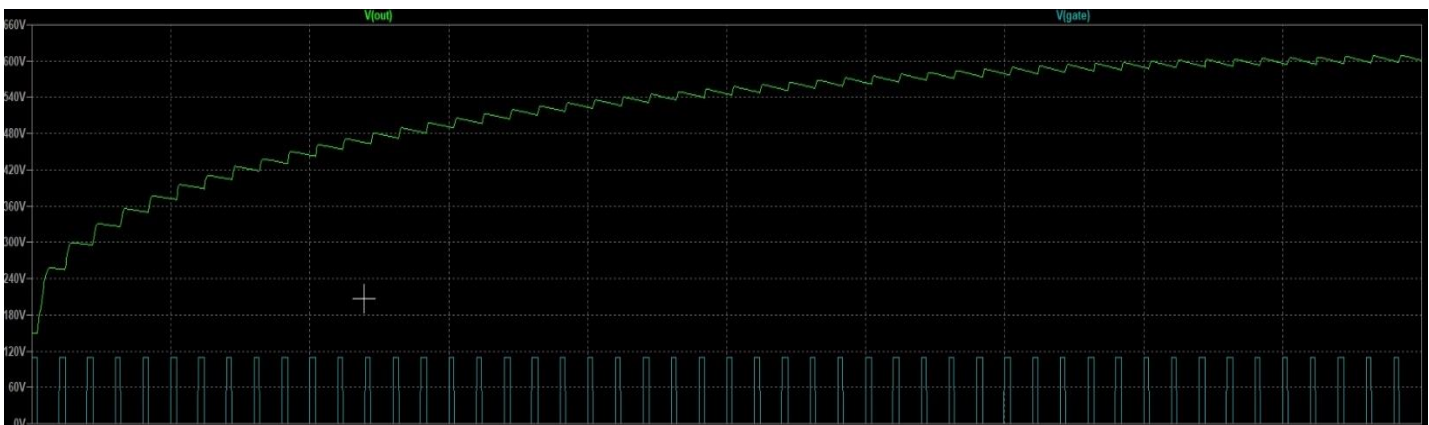


Figure 17: gate voltage in (turquoise) and output voltage(green)

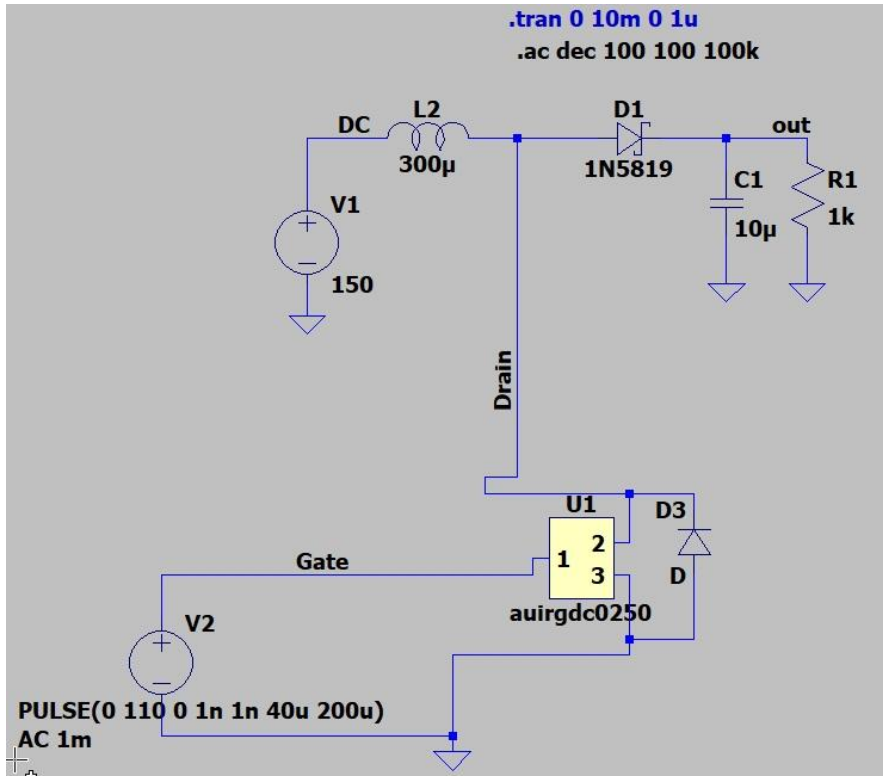


Figure 18: IGBT implementation in Open loop Boost converter

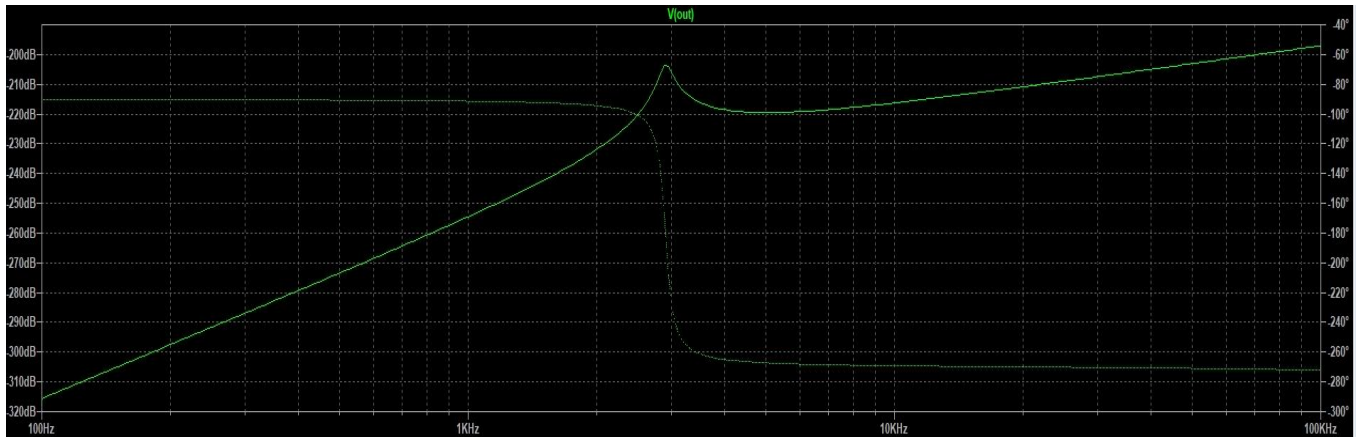


Figure 19: Transfer function of the IGBT implementation in Open loop Boost converter

- (c) Add 0.1 ohm of the equivalent series resistance (ESR) to the inductor. Plot the input, output, voltage, current, and input gate voltage of the upper switch. In this simulation, use ideal capacitor without any resistance. Estimate the total converter efficiency.

|        | Input voltage (RMS) | Output voltage (RMS) | Input current (RMS) | output current (RMS) | Duty ratio [%] | Efficiency [%] |
|--------|---------------------|----------------------|---------------------|----------------------|----------------|----------------|
| Values | 150V                | 521.03V              | 6.288A              | 521.03mA             | 0.72           | 15%            |

Adding inductor internal resistance  $ESR = 0.1\Omega$  in series will significantly reduce the output current.

Plots:

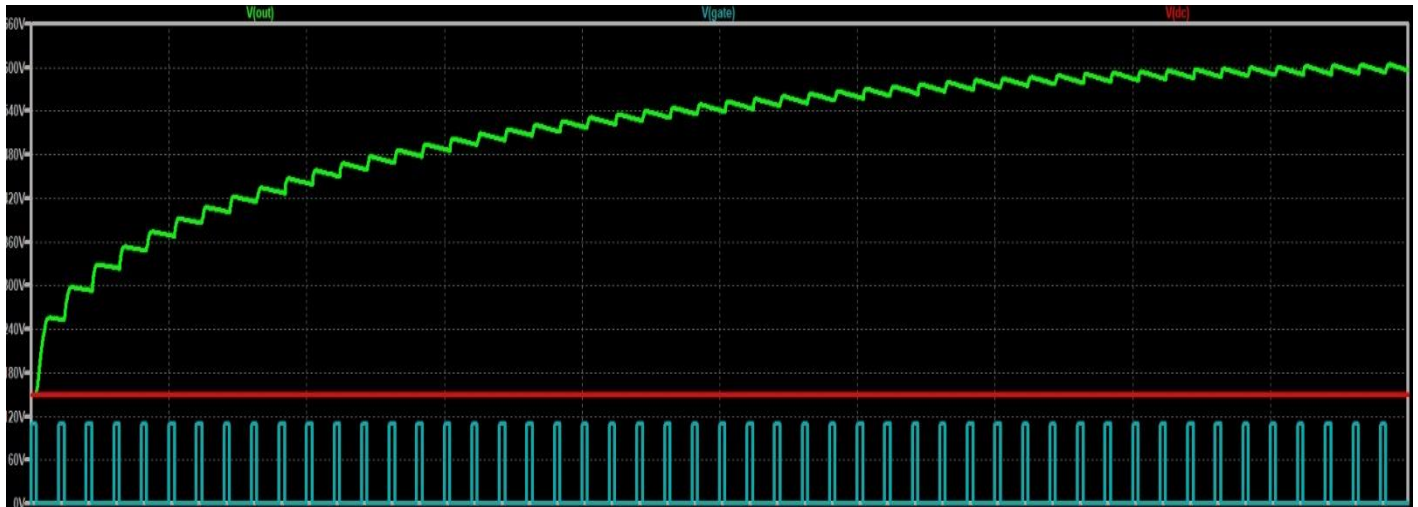


Figure 21: Boost converter output Voltage goes up to 600V(green), input voltage(red), Input gate voltage(turquoise)

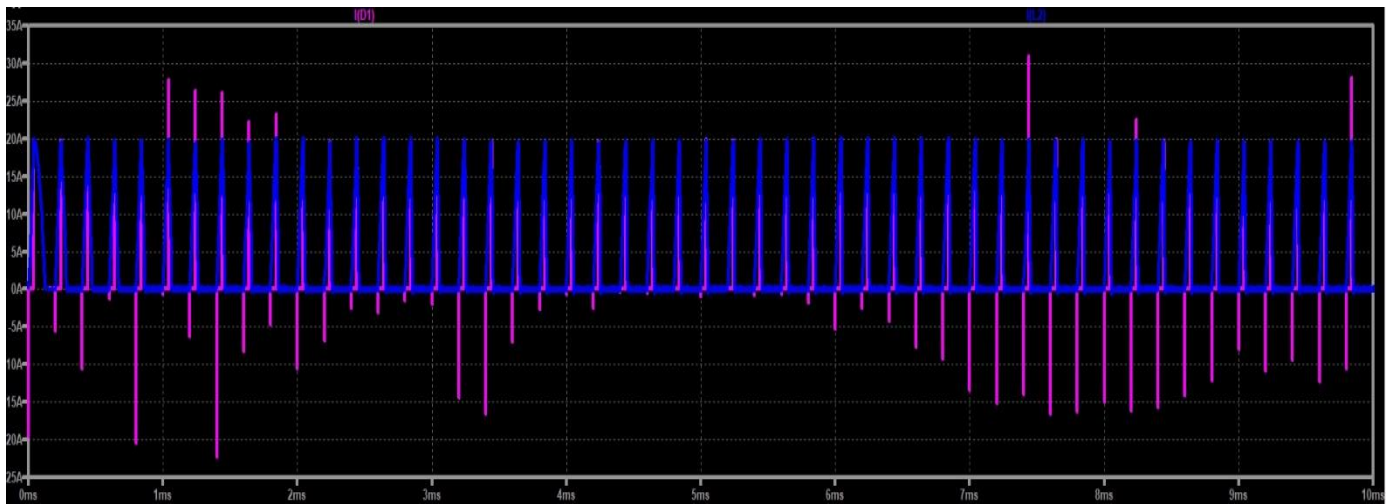


Figure 20: current through the inductor(blue), diode(magenta), output resistor(grey)

(d) Calculate switching losses (turn on and off) and compare them with simulation results.

| Losses        | Calculated value | Simulation result      |
|---------------|------------------|------------------------|
| Turn on loss  | 0.04266 Watts    | 61.066 $\mu\text{J/S}$ |
| Turn off loss | 0.033 Watts      | 58.756 $\mu\text{J/S}$ |

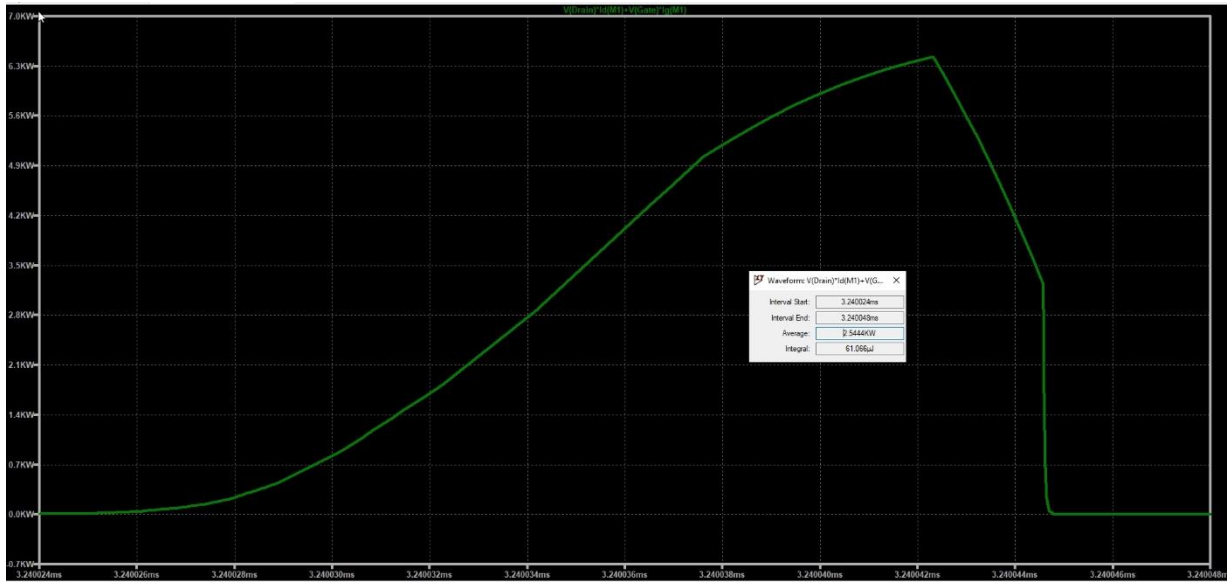


Figure 22; MOSFET turn-ON loss plot.

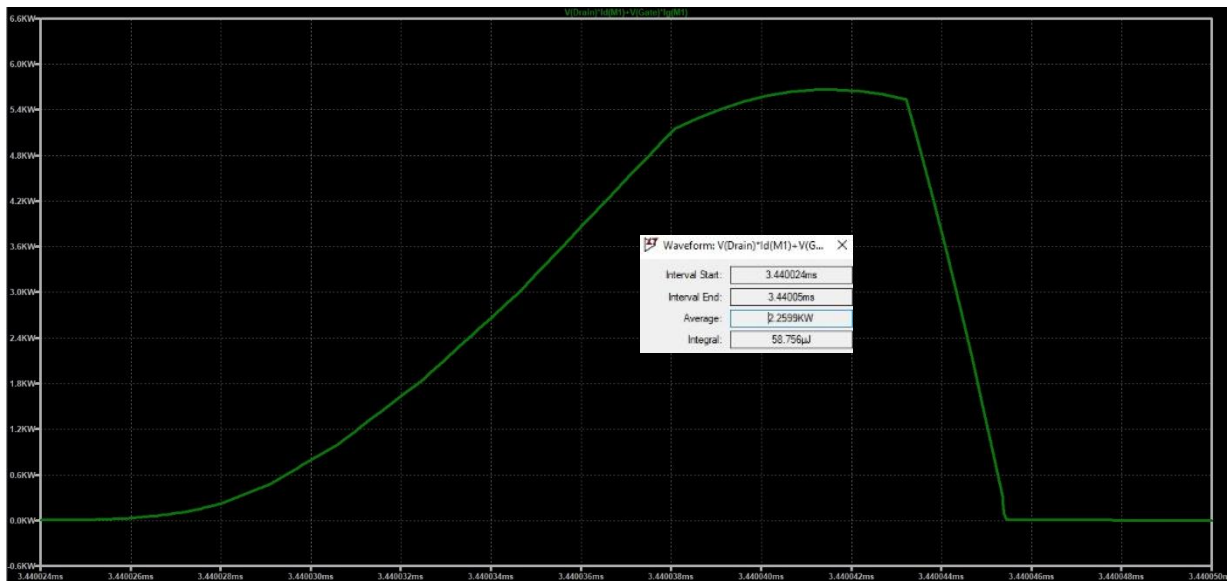


Figure 23:Figure 7: MOSFET turn-OFF loss plot.58.756 $\mu\text{J}$

Below is the closed loop circuit schematic and transfer function of the Boost converter:



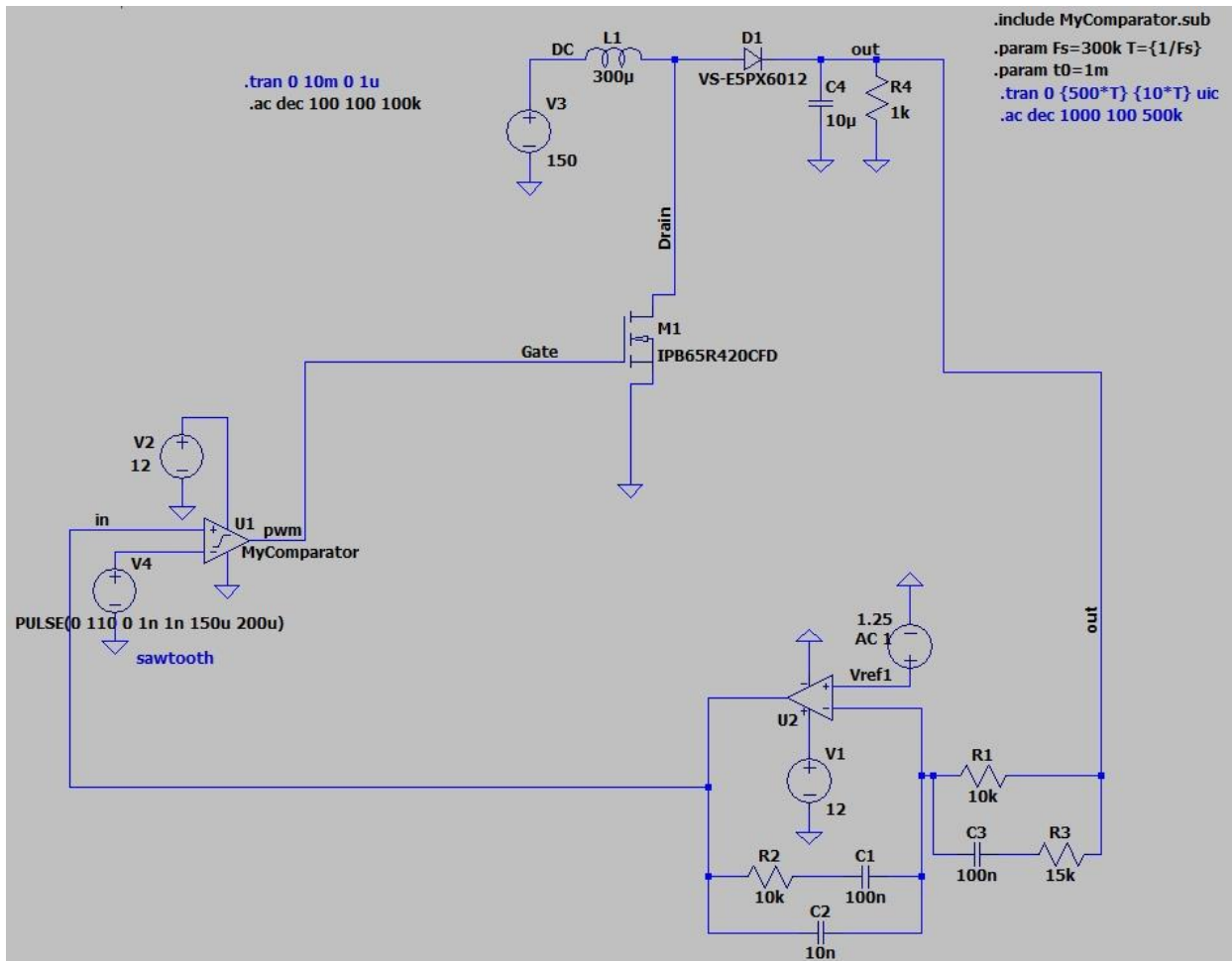


Figure 24: Boost converter closed loop circuit

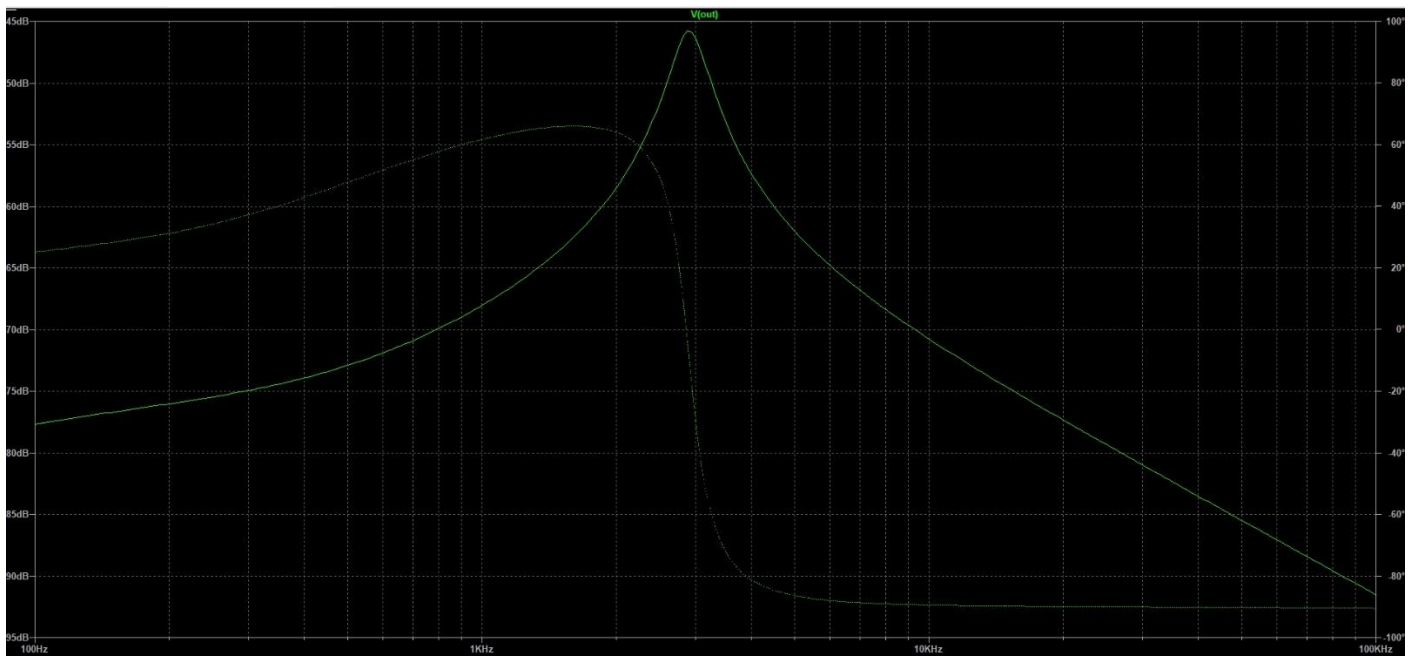


Figure 25: Boost converter closed loop transfer function phase and magnitude.



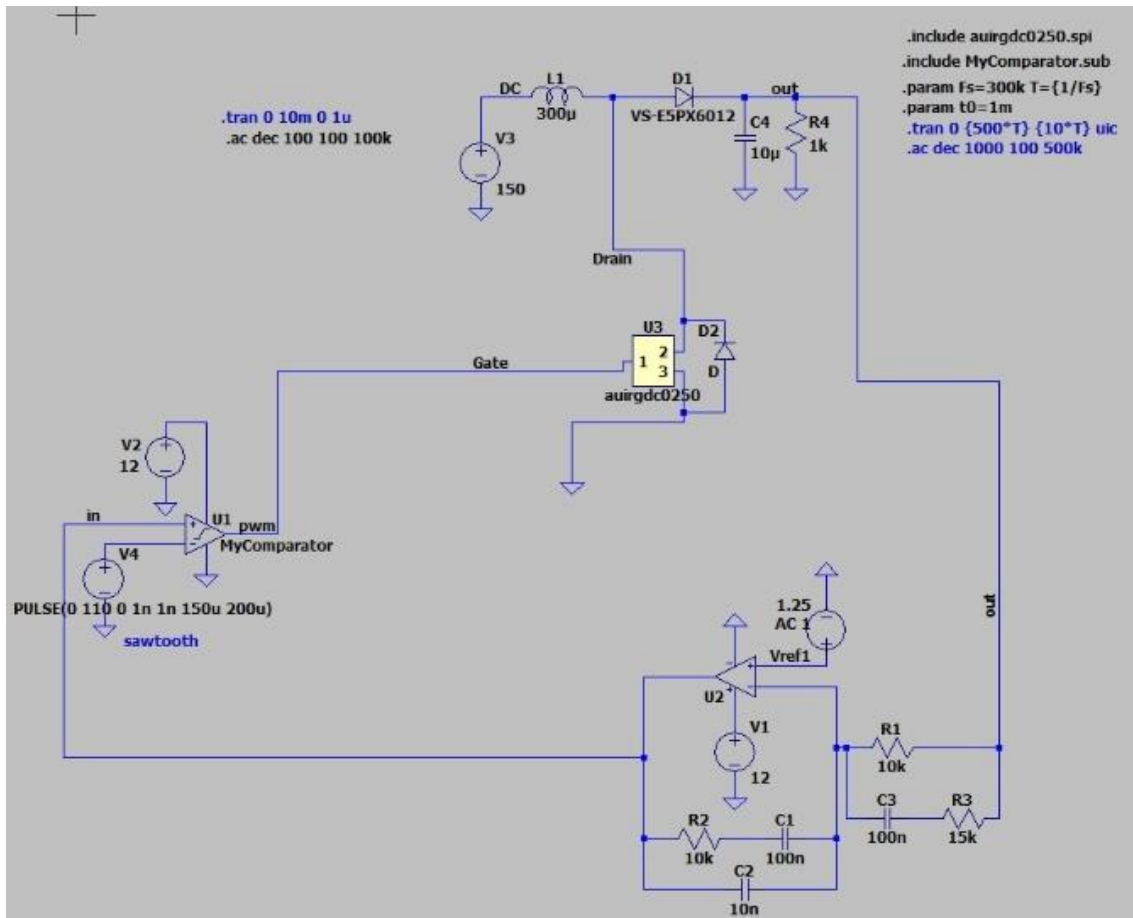


Figure 26:IGBT implementation in Boost converter closed loop

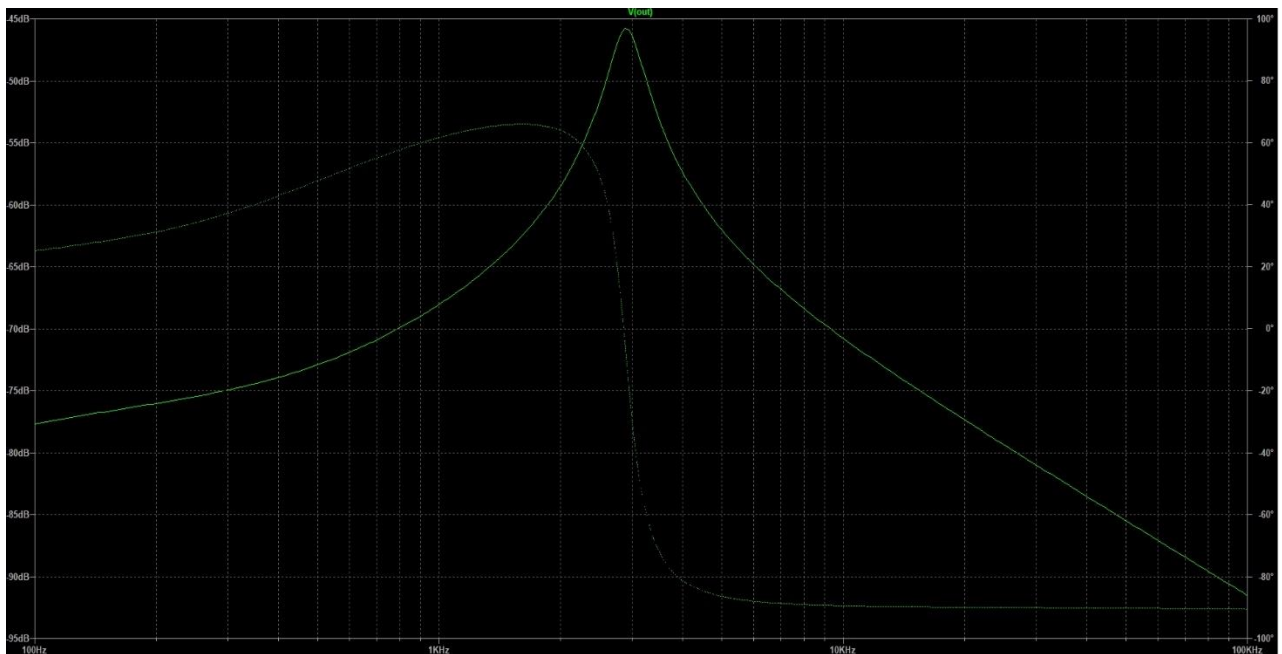


Figure 27:Transfer function of the above closed loop Boost converter.

**Appendix-A**

The following resources are collected from the PURDUE UNIVERSITY course lecture materials for:

*ECE 61016 - Power Electronic Converters and Systems*

*Instructor: Woongkul (Matt) Lee*

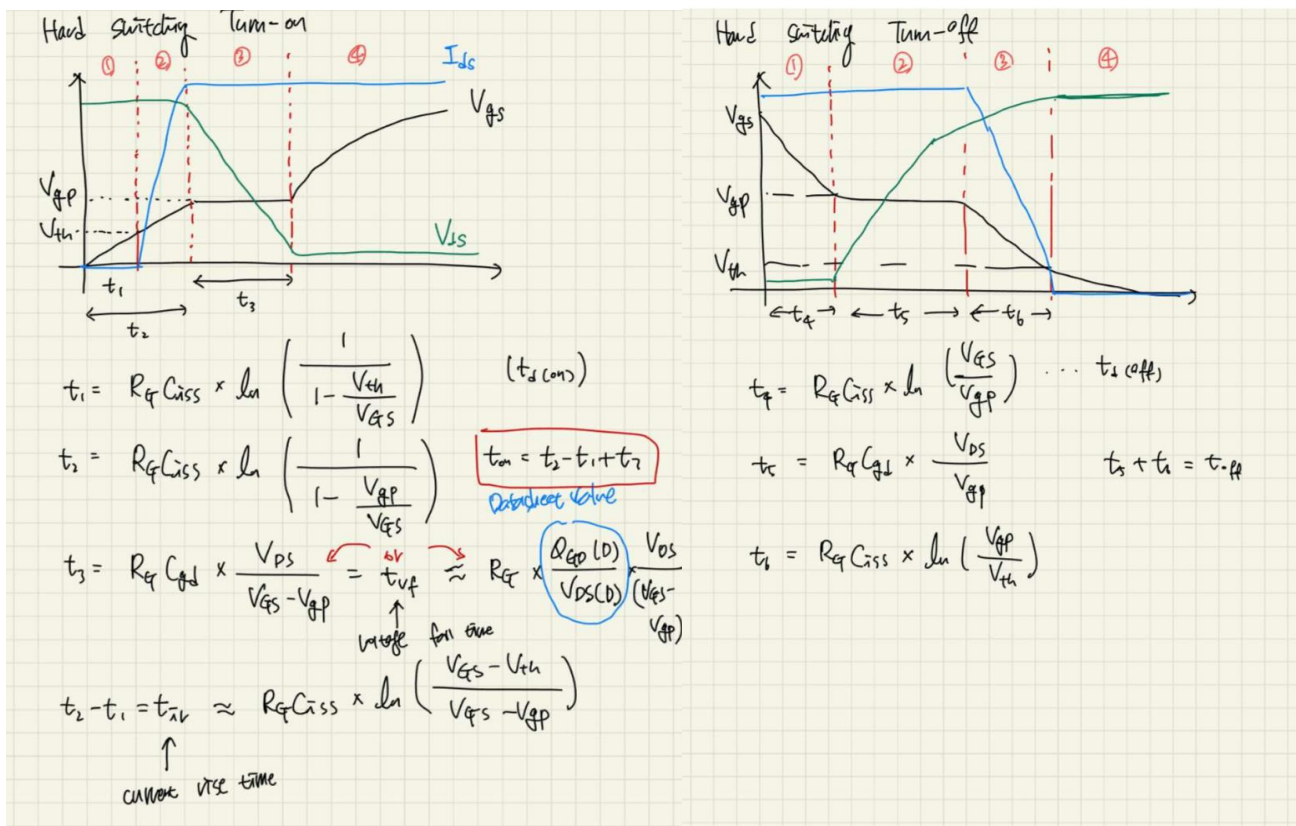
*Assistant Professor*

*Elmore Family School of Electrical and Computer Engineering*

*Purdue University*

*West Lafayette, IN*

*Seng-Liang Wang Hall, Room 2061*



$$V_{GS} = 10V$$

$$V_{TH} = 2.2V @ 25^\circ C \text{ (from } V_{GS} \text{ vs. } I_D \text{ graph)}$$

$$V_{GP} = 2.5V \text{ (from } \theta_{gs} \text{ vs. } V_{GS} \text{ graph)}$$

$$R_{GE} = 1\Omega, 350\Omega$$

$$R_{g(int)} \approx 1\Omega$$

• now,

$$t_d(on) = t_1 = R_{GE} C_{iss} \times \ln \left[ \frac{1}{1 - \frac{V_{TH}}{V_{GS}}} \right]$$

Forc.  $R_{GE} = 1\Omega$  (datasheet value)  $\rightarrow t_d(on) = 0.893 \text{ ns}$

Forc.  $R_{GE} = 350\Omega$   $\rightarrow t_d(on) = 313 \text{ ns}$

•  $t_{on} = t_2 - t_1 + t_3 = t_{irc} + t_3$

$$= \left[ R_{GE} C_{iss} \times \ln \left\{ \frac{V_{GS} - V_{TH}}{V_{GS} - V_{GP}} \right\} \right]$$

$$+ \left[ R_{GE} \frac{\theta_{gs(D)}}{V_{DS(D)}} \times \frac{V_{DS}}{(V_{GS} - V_{GP})} \right]$$

if,  $V_{DS} = 12V$ ;  $\theta_{gs} \approx 4 \text{ nC}$   
 forc,  $V_{DS(D)} = 15V$   
 $V_{GS} = 4.5V$

Then,  $t_{on} \approx 830 \text{ ns}$

now, if,  $V_{DS} = 12V$ ;  $\theta_{gs} \approx 4 \text{ nC}$ ,  $V_{DS(D)} = 15V$   
 $V_{GS} = 4.5V$

Then,

$$t_{on} = 2.37 \text{ ns}$$

i/p cap.

$$C_{iss} \approx 3525 \text{ PF (@ } V_{GS} = 0V)$$

O/P cap.

$$C_{oss} \approx 1040 \text{ PF (@ } V_{GS} = 0V)$$

Reverse Transfer cap.

$$C_{rss} \approx 79 \text{ PF (@ } V_{GS} = 0V)$$

$$C_{rss}/C_{iss} \text{ ratio} \approx 0.0222 \text{ PF (@ } V_{GS} = 0V)$$

$$R_G = 1 \text{ k}\Omega$$

$$\begin{aligned} \bullet t_d(\text{off}) &= R_G C_{iss} \times \ln\left(\frac{V_{GS}}{V_{GP}}\right) \\ &= 4.84 \text{ ns} \end{aligned}$$

$$R_G = 350 \text{ k}\Omega$$

$$\bullet t_d(\text{off}) = 1695 \text{ ns}$$

$$\begin{aligned} \bullet t_{off} &= t_s + t_b = R_G C_{gd} \times \frac{V_{DS}}{V_{GP}} \\ &\quad + R_G C_{iss} \times \ln\left(\frac{V_{GP}}{V_{TH}}\right) \end{aligned}$$

$$\Rightarrow R_G \times \frac{g_{m(D)}}{V_{DS(D)}} \times \frac{V_{DS}}{V_{GP}} +$$

$$R_G C_{iss} \times \ln\left(\frac{V_{GP}}{V_{TH}}\right)$$

$$\Rightarrow 1.83 \text{ ns}$$

$$t_{off} = 640 \text{ ns}$$

$$\bullet E_{on} = \frac{1}{2} V_{DS} \times I_D \times t_{on}$$

$$\text{Here, } I_D = 10 \text{ A}$$

$$t_{on} = 2.37 \text{ ns}$$

$$V_{DS} = 12 \text{ V}$$

$$\text{So, } E_{on} = 142.2 \text{ nJ}$$

$$\bullet E_{on} \text{ for } R_G = 350 \text{ k}\Omega$$

$$E_{on} = 142.2 \text{ nJ}$$

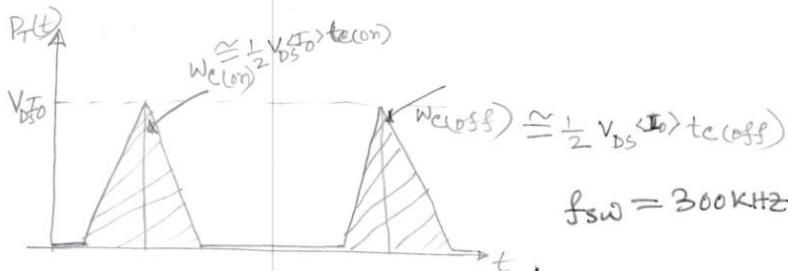
$$\bullet E_{off} = E_{on} = 142.2 \text{ nJ}$$

$$\bullet E_{off} = E_{on} = 142.2 \text{ nJ}$$

(b)

| switching times      | calculated values                                    | datasheet values |
|----------------------|------------------------------------------------------|------------------|
| $t_{d(on)} = t_1$    | $0.893 \text{ ns (Forc. } R_{GE} = 1 \Omega)$        | 12-24 ns         |
| $t_{rr} = t_3$       | $1.68 \text{ ns (Forc. } R_{GE} = 1 \Omega)$         | 10-20 ns         |
| $t_{d(off)}$         | $4.84 \text{ ns (Forc. } R_{GE} = 1 \Omega)$         | 30-60 ns         |
| $t_f = t_{rr} = t_5$ | $\approx 1.83 \text{ ns (Forc. } R_{GE} = 1 \Omega)$ | 10-20 ns         |

(c) total switching loss of top switch (MOSFET)



$$f_{sw} = 300 \text{ kHz}$$

$$\begin{aligned}
 W_{c(on)} &\approx \frac{1}{2} V_{ds} I_D t_{don} \times f_{sw} \\
 &= \frac{1}{2} \times 12 \times 10 \times 2.37 \text{ ns} \left[ \begin{array}{l} \text{Forc.} \\ 300 \text{ kHz} \end{array} \right] \\
 &\quad \left[ \begin{array}{l} \text{Forc.} \\ R_{GE} = 1 \Omega \end{array} \right] \\
 &= 0.04266 \text{ watts or, J/s}
 \end{aligned}$$

$$\begin{aligned}
 W_{c(off)} &\approx \frac{1}{2} V_{ds} I_D t_{do} \times f_{sw} \\
 &= \frac{1}{2} \times 12 \times 10 \times 1.83 \text{ ns} \times 300 \text{ kHz} \\
 &= 0.033 \text{ watts or, J/s} \left[ \begin{array}{l} \text{Forc.} \\ R_{GE} = 1 \Omega \end{array} \right]
 \end{aligned}$$

$$\text{Forc. } R_{GE} = 350 \Omega$$

$$\begin{aligned}
 W_{c(on)} &= \frac{1}{2} \times 12 \times 10 \times 830 \text{ ns} \times f_{sw} \\
 &= 14.94 \text{ watts}
 \end{aligned}$$

$$\begin{aligned}
 W_{c(off)} &= \frac{1}{2} \times 12 \times 10 \times 640 \text{ ns} \times f_{sw} \\
 &= 11.52 \text{ watts}
 \end{aligned}$$

## References:

1. <https://www.vishay.com/docs/73152/si7336adp.pdf>
2. [https://www.infineon.com/dgdl/Infineon-BSC060N10NS3%20G-DataSheet-v02\\_05-EN.pdf?fileId=db3a30431ce5fb52011d1aab7f90133a](https://www.infineon.com/dgdl/Infineon-BSC060N10NS3%20G-DataSheet-v02_05-EN.pdf?fileId=db3a30431ce5fb52011d1aab7f90133a)



